

The Big Data Bandits

Video Game Trends

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Provide Team Members' Names and UIS Emails

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Abstract and Highlights

The entertainment industry is one of the largest in the world, and video games make a significant part of it alongside movies, music and other media. Many big video game companies such as Nintendo, Sony, Tencent, SEGA, Microsoft, etc. spend millions of dollars in producing a video game, and there is a big chance of hit or miss (whether that video game will attract various customers). The three big markets of the world for video games are North America Region, Europe Region, and Japan region. Our project will attempt to explore and analyze if there is any relationship between gaming platform, genre, and sales, as we try to see the patterns in analyzing the video game sales data which has been compiled over the years. We hope our model will be able to help those game publishers decide which genre to target in a specific market region and reduce the risk of loss in sales.

Our research focus on these questions:

1. Popularity of Games by genre over the years from 2000 to 2016?
2. Which genre is the most popular over the years in North America region?
3. Which platform is the most popular over the years in North America region?
4. Using unsupervised learning, which genre clusters of games are the most similar in characteristics such as platform and sales?
5. Given the estimated targeted sales of different regions, and platform, which genre should the publisher produce?

For the project, we have developed and deployed various models for both exploratory and predictive purposes. We have used SAS, Tableau, and excel to execute and develop our models. We have done cluster analysis, using ward clustering, centroid and average clustering, this was done in SAS for exploratory purposes. For the clusters both automatic and user-specified methods were used. For performance evaluation, we have evaluated the stability of the clusters by checking the number of clusters before and after sampling. For the predictive part, we have developed and executed three models in the SAS, them being, decision tree, regression, and neural network. We have developed multiple versions for each of the above-mentioned supervised data mining techniques in order to get the best model possible. We then compared the best model from the three supervised datamining techniques to arrive at the best model. We used the fit statistics to evaluate the performance of the above models.

We have scored the best model on our score data set to find the predictive power of our model. Some of the recommendations from the analysis of data we have done is to focus more on handheld devices in the Japan market and more focus on bigger consoles for the North American Market

2. Problem Description

Description and some background about the focal problem addressed in the project:

In this project we aim to analyze the video game dataset, which has been sourced from Kaggle. The main problem addressed in this project is to see the relationship between video game sales, with respect to geographical location (meaning continental location, Europe, North America or Japan), genre of the games sold, platform where the game is hosted.

We also want to explore the relationship between the geographical location of the video game consumers and platform where the game is hosted.

Why it is an important problem to be addressed

It is an important problem to be analyzed for the following reasons:

Knowing your target market is key in any business endeavor, by knowing the target market the marketers can do a better job of developing the game and executing a successful marketing campaign based on platform, region and genre.

There is much more to it than the commercial aspect, the type of video games consumers like is also indicative of the psychological state of affairs of a population. It shows whether they have a higher propensity to be violent when the conditions are right, whether they are more analytical, philosophical etc.

Similarly there have been studies showing that people who ride SUV's are more obsessed with status, less likely to volunteer, No strong connection to their community, less giving, more afraid of crime, more likely to text and drive, and more risk taking when driving. (Bradsher, 2003).

This data hence can be very useful to perform studies in other disciplines such as psychology, sociology etc.

Reference:

Bradsher, K. (2003). *High and mighty: The dangerous rise of the SUV*. Public Affairs.

Well-articulated question(s) for predictive and exploratory analysis

Predictive Analysis:

Predict what type of games will be more likely to be sold in a particular region

Exploratory Analysis:

Explore the relationship between Genre, Region

Technical/analytical Methodology

For our research, the data we use is about video game sales from around the world. The dataset has been used in a multitude of studies as the data is holistic and has potential to show relationships between regional sales (region being continents), platform of the video game, publishers.

The source of the data is Kaggle. The dataset has been compiled from various sources, one of the significant ones being from VgChartz.com. The link for the dataset is <https://www.kaggle.com/datasets/gregorut/videogamesales>

For our tools, we use excel for cleaning, SAS for analysis, and tableau for visualizing.

Before we start preparing our data, we clean our data in Excel to ensure the accuracy and reliability of our dataset. This includes handling missing values, removing duplicates, and correcting any inconsistencies in the data. Once we are pass this stage, we convert the excel file into SAS file for exploratory modeling and use the copy of that SAS dataset file to further divide into two data sets using SAS for scoring and modeling, for predictive analysis.

We then load the original excel data set into tableau to answer Q1, Q2, and Q3.

We then use SAS to answer the remaining questions. In the exploratory analysis phase, we employed major clustering techniques such as ward clustering, average clustering, and centroid clustering. These techniques help identify patterns and group similar data points together, providing valuable insights into the underlying structure of the dataset.

To improve the quality of our clustering results, we repeated the clustering process on a 90% sample of the data. By applying the same clustering techniques multiple times, we aimed to enhance the accuracy and robustness of our clustering results.

For predictive analysis, we employed regression, decision trees, and neural networks. These techniques allow us to make predictions and uncover relationships between variables in the dataset. By comparing different models, we generated a comprehensive score report that enables further analysis and decision-making.

Our project emphasizes both exploratory and predictive analysis to provide a holistic understanding of the dataset. The combination of these approaches allows us to uncover insights, discover trends, and make accurate predictions in the gaming industry.

The project demonstrates a systematic and rigorous approach to data analysis, combining data cleaning, visualization, clustering, and predictive modeling techniques. This ensures that the insights and conclusions drawn from the analysis are based on robust methodologies and reliable data.

Data cleaning:

Often in data mining the majority of the time is not spent in developing and deploying models, a big chunk of time is taken to clean the data. As data often does not come in a ready to use format, often times data miners need to change the format of data, remove duplicates as it can often skew the results because of duplicates.

In our case for the video games dataset fortunately the data was complete, but just had some duplicates to be removed, which we have removed.

By doing this step we have ensure that the data is accurate, non-skewed and ready to be deployed.

Data Visualization to explore the data set (Using Graphs):

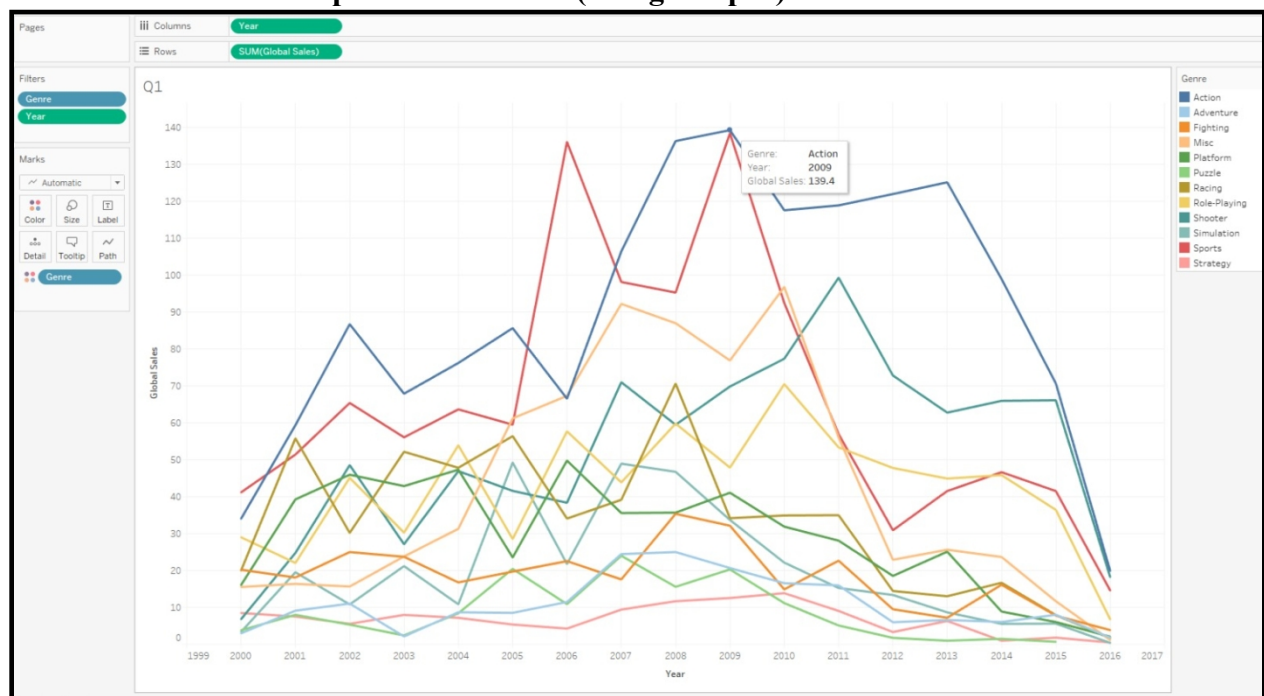


Figure 1 Global Sales of Video Games by genre over the years (2000 – 2016)

The above figure shows the video game sales by different genres over time from 2000 to 2016. The legend shows the genre with different color coding. We can interpret from the graph that the highest global sales are from the action and sports category. The least sales are from adventure, strategy, and puzzle.

The rest of the genres are hovering in between the highest and lowest video game sales.

We can see the trend of video game global sales increasing over time, this also makes intuitive sense, as computing power, affordability of purchasing computing and gaming devices has increased vastly over time.

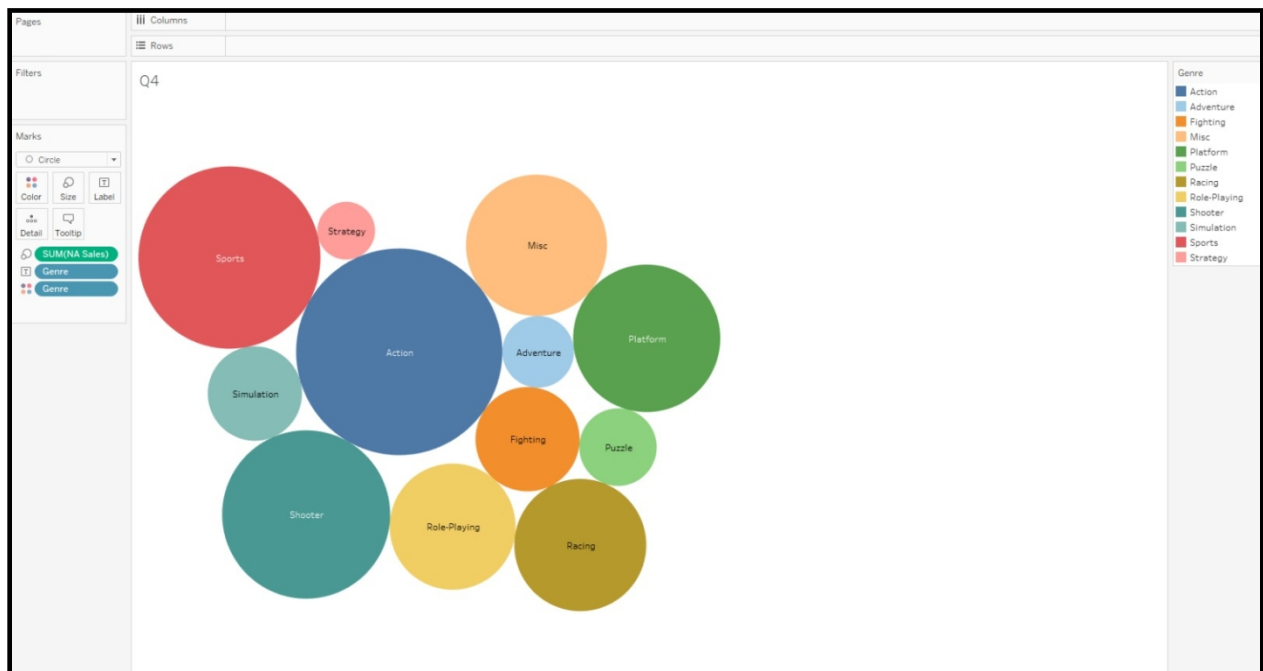


Figure 2 Sales of video game by genres in North America

Figure 2 is about the sales of video games by genres in North America. The legend in the top right corner shows the various categories, while the size of the circles shows the relative proportion of sales. We can infer from the above figure, that the highest sales in North America are usually games with action genre, closely followed by sports genre.

Strategy, adventure and genre have the least number of sales in North America. This may hint at an idea that NA region favors more action-packed genre games rather than other categories, and games of such nature would sell well for this region.



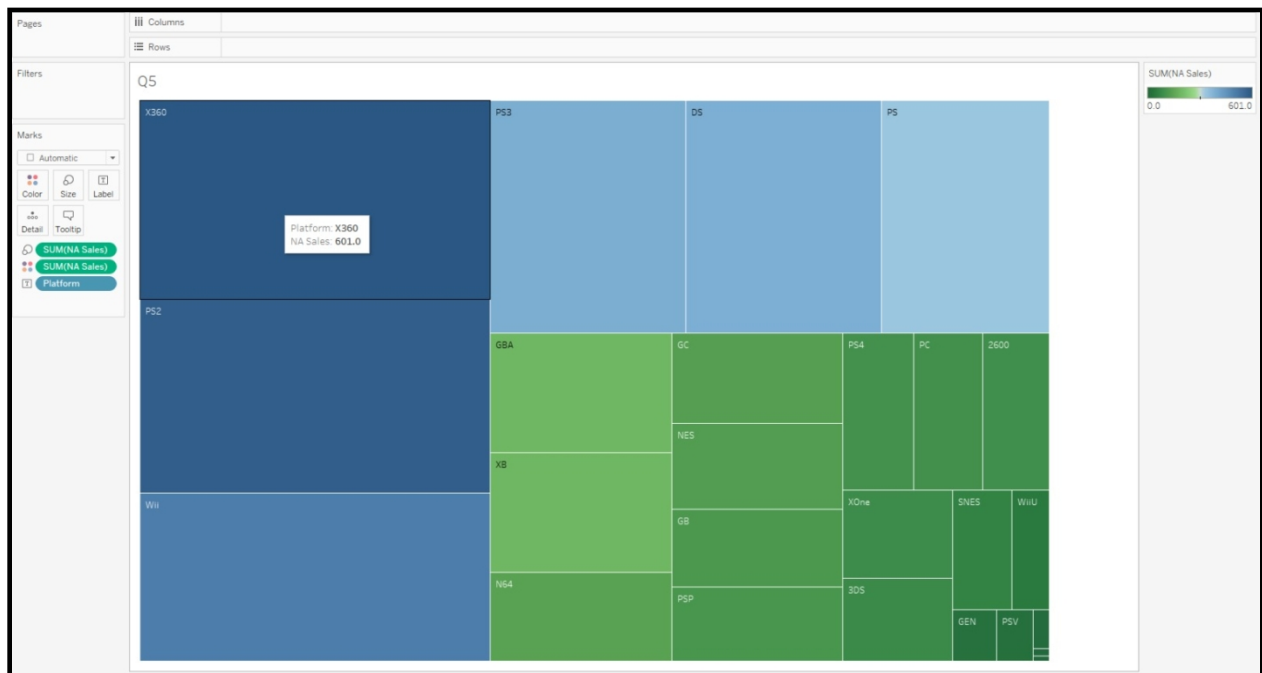


Figure 3 Sales of video games by platform in North America

The above figure 3 shows the sale of video games by platform in North America. The above diagram shows that the highest sales is of platform X360 (Xbox 360) in North America, closely followed by PS2, Wii, PS3, DS, PS. The least sales being of platforms GEN, PSV, 3DS. This hints that NA region consumers may favor more towards non-mobile consoles that pack more power (graphics) instead of mobility.

Exploratory Analysis:

Exploratory analysis is an unsupervised data mining technique which utilizes clustering methods to find patterns and group the data. In unsupervised data mining technique, we do not have any target variable. It groups the clusters together based on similarity.

For our project we have used all the major clustering algorithms: Ward clustering, Centroid Clustering and Average clustering.

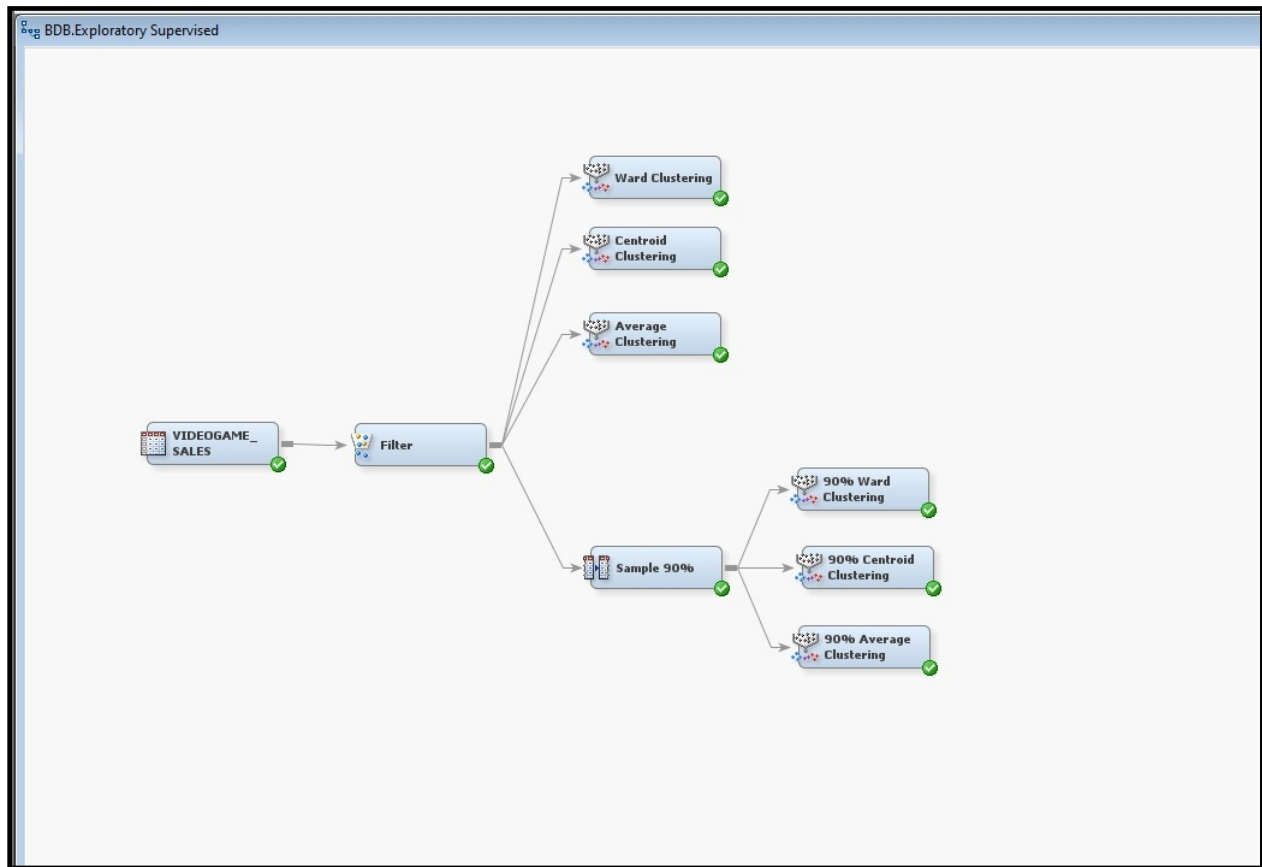


Figure 4: Exploratory XML Diagram

In clustering there are two types, K-Means clustering and hierarchical clustering. Hierarchical clustering is unsupervised in that the number of clusters are created automatically. In hierarchical clustering we have got 20 clusters for Ward clustering, 11 clusters for centroid clustering and 7 for average clustering.

For the 90% sample, we have got 22 clusters for 90% ward clustering, 10 for 90% centroid clustering and 5 for 90% average clustering.

We can infer from the above that the number of clusters before and after sampling for all clustering algorithms (ward, centroid and average clustering) are within ± 2 range. But the number of clusters are greater than 6 this makes it harder to interpret.

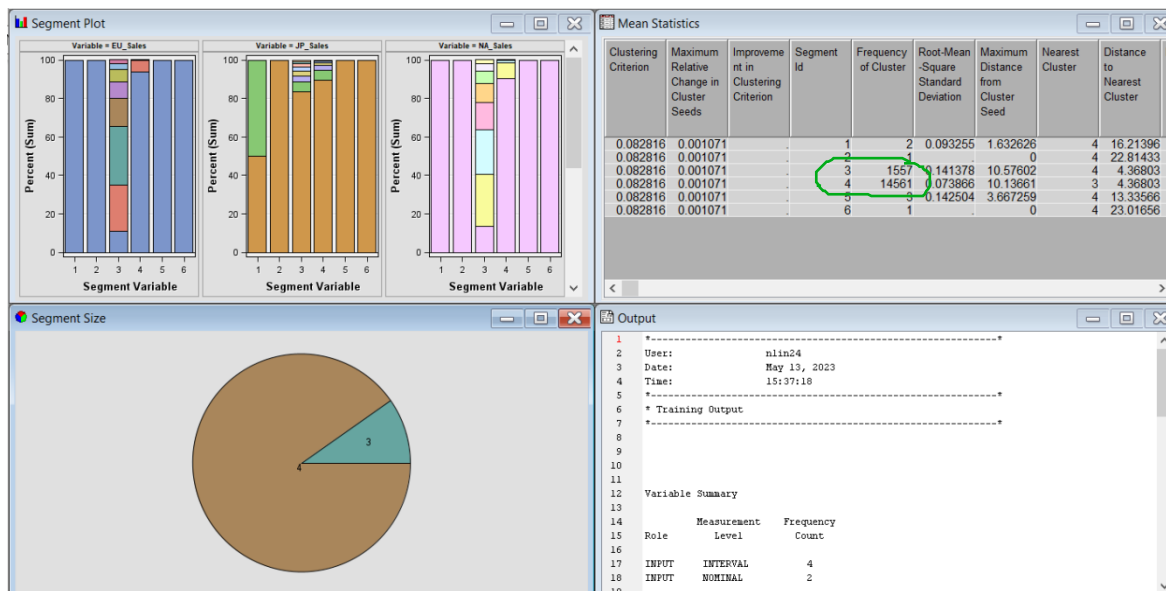
If the number of clusters are greater than 6 for any given clustering algorithm, it is harder to interpret the results of clustering and generate meaning insights. When we did hierarchical clustering, we found that the number of clusters for all our algorithms was above six, so we then proceeded to do K-means clustering by limiting the number of clusters to 5 for ward, centroid and average clustering. (K. P. Sinaga and M.S. Yang, 2020)

Variable Name	Label	Number of Splitting Rules	Number of Surrogate Rules	Importance
NA_Sales		7	6	1
Other_Sales		6	7	0.968798
EU_Sales		4	8	0.967207
Platform		0	7	0.298795
JP_Sales		1	5	0.223776
Publisher		0	2	0.165653

From the results of ward clustering window. Going to the view tab, than cluster profile and variable importance we get the above result.

It shows that the most important variable is North American sales data.

We thought about pruning the data, and combining the game genres, however, the nature of game genres differs and merging them for the sake of pruning would give us an inaccurate model. We tried lowering the number of clusters to 6, to make it more manageable, and the results gave us two big clusters that most games can be classified into based on their similarity.

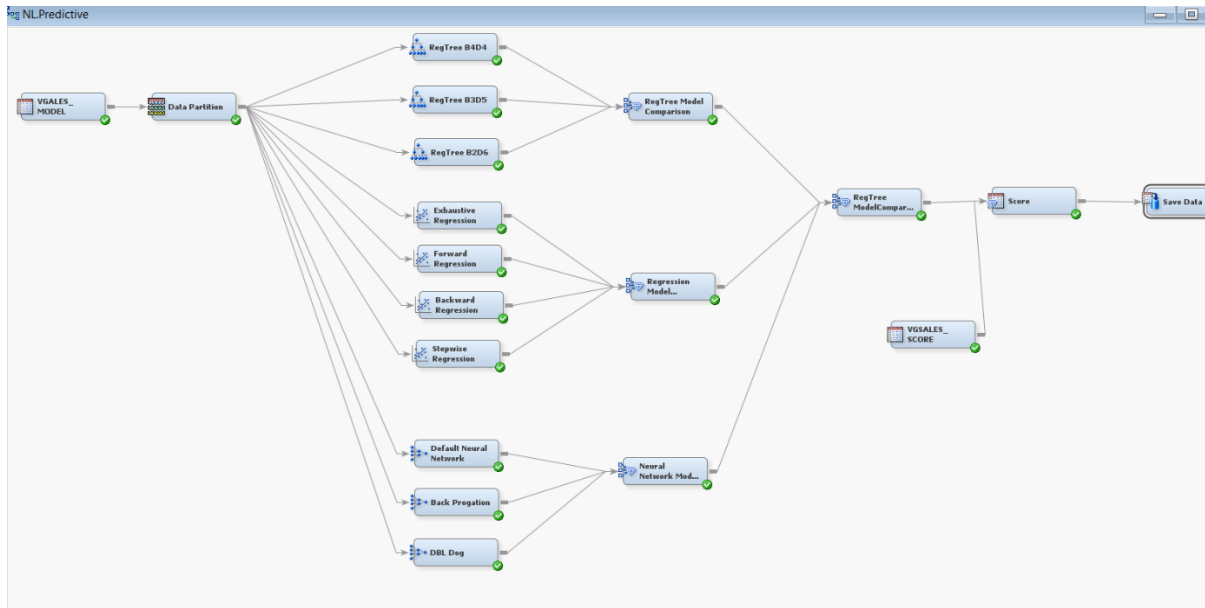


Based on this exploratory analysis, and focusing on NA sales in pink, most games would fall into cluster 3 or cluster 4 similarities. This tells us that new games that will be produced with similar characteristics, will usually fall into those two clusters, and based off that similar characteristics in that cluster, we can have an idea of how the sales of that genre in those games would perform based on other similar games in that category.

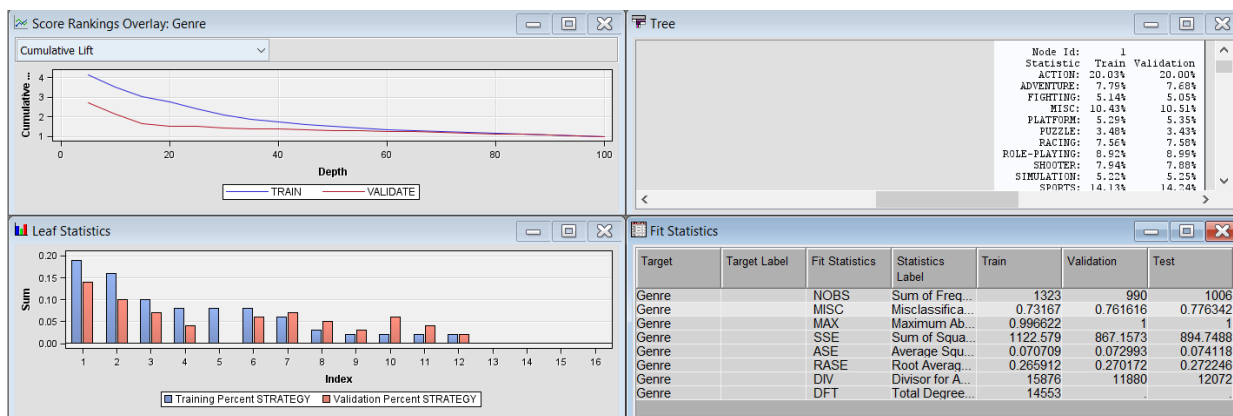
Predictive Modeling

For our predictive modeling part, we have developed and deployed all the three major supervised data mining techniques: decision trees, logistic regression and neural nets. We use the data set that has been divided into model and score purposes.

Below is the overall structure of our predictive analysis

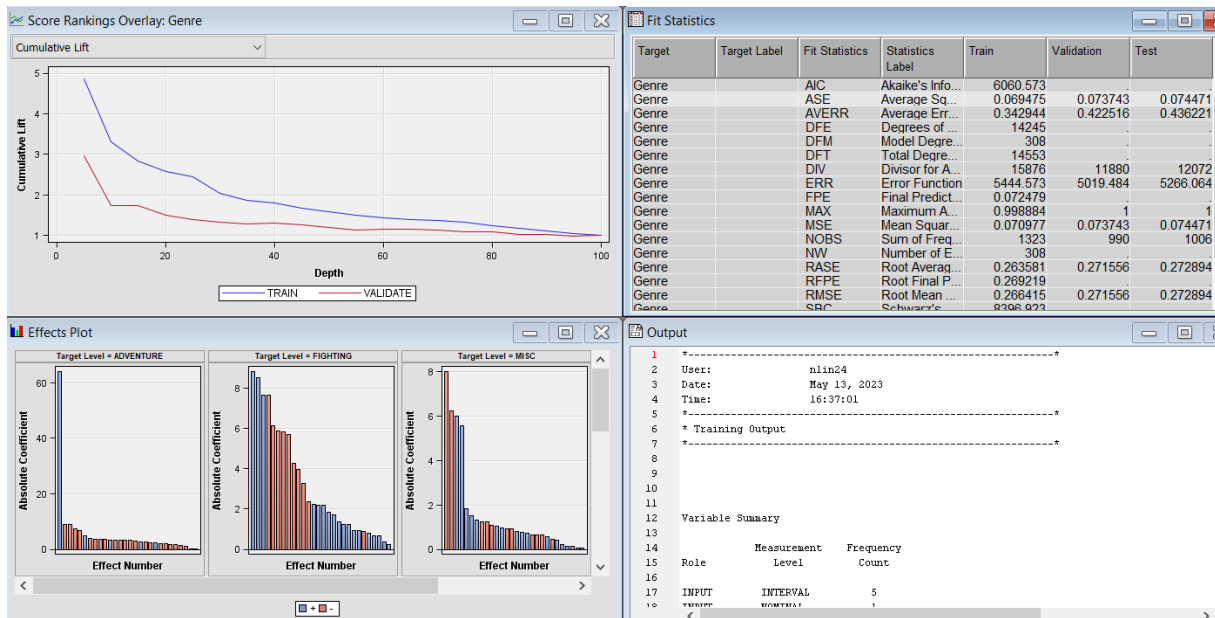


For decision trees, we use models with breadth 4, depth 4; Breadth 3, depth 5; and breadth 2, depth 6. The best model for this came out to be B4D4, through the model comparison node. Evaluation for this is checking the accuracy and misclassification rate of the model.

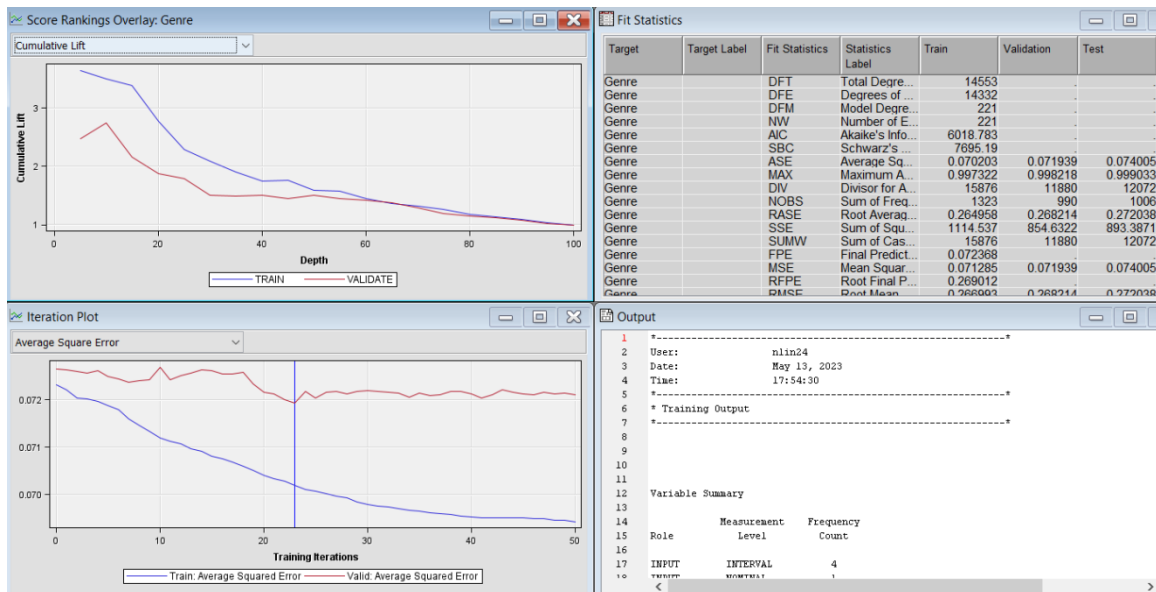


For the logistics regression model, we use Exhaustive regression, Forward Regression, Backward Regression and Stepwise Regression. The best model from the logistics regression models came out to be the forward regression. The performance evaluation is done by looking at average

squared error and root average squared error and the best model consists of the least error in both of those metrics.

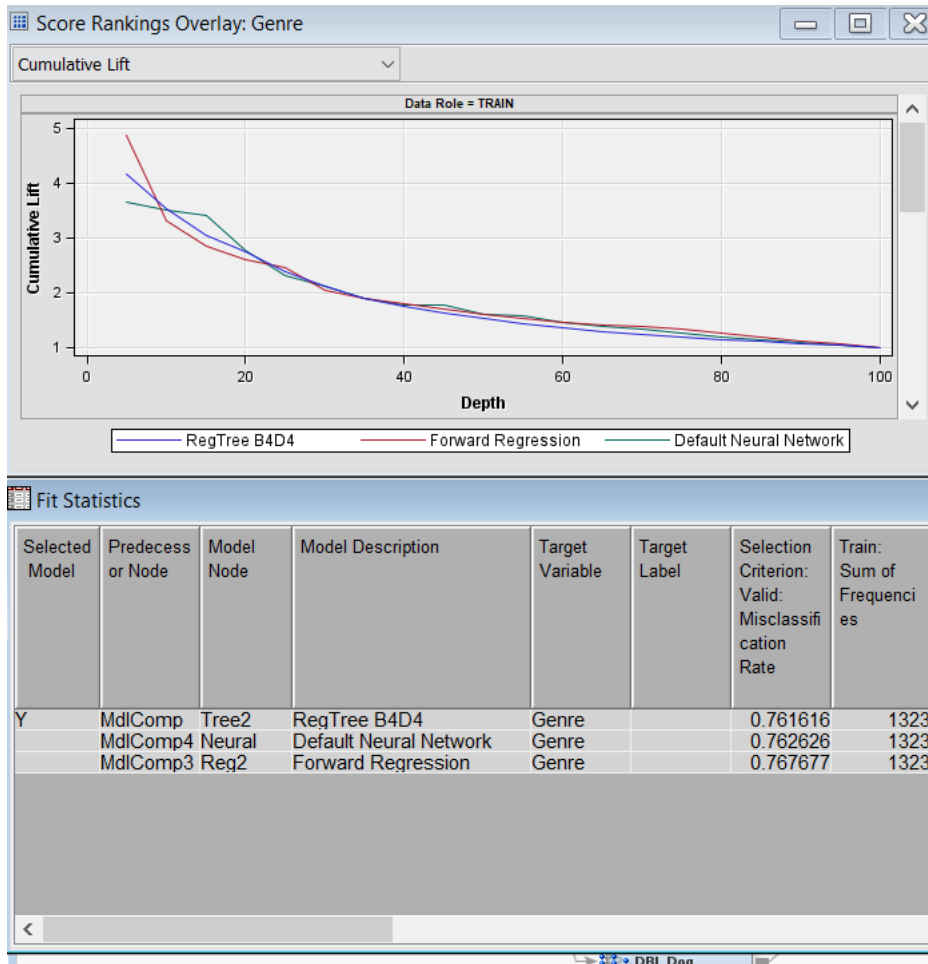


For Neural networks, we use default NN, Back propagation NN, and DB Dog NN, with 5 hidden layers assigned. The best performance model resulted in DBL Dog NN. The detailed evaluation of how the NN is a Blackbox, and we do not have enough information on how the detailed scope of how it works.



We then compare the three best model nodes, and the best node chosen by SAS turns out to be Regression Tree B4D4. Decision trees are well known for its usability in multiple disciplines and ease of interpretation. (Song & Lu, 2015)

Here are the results:



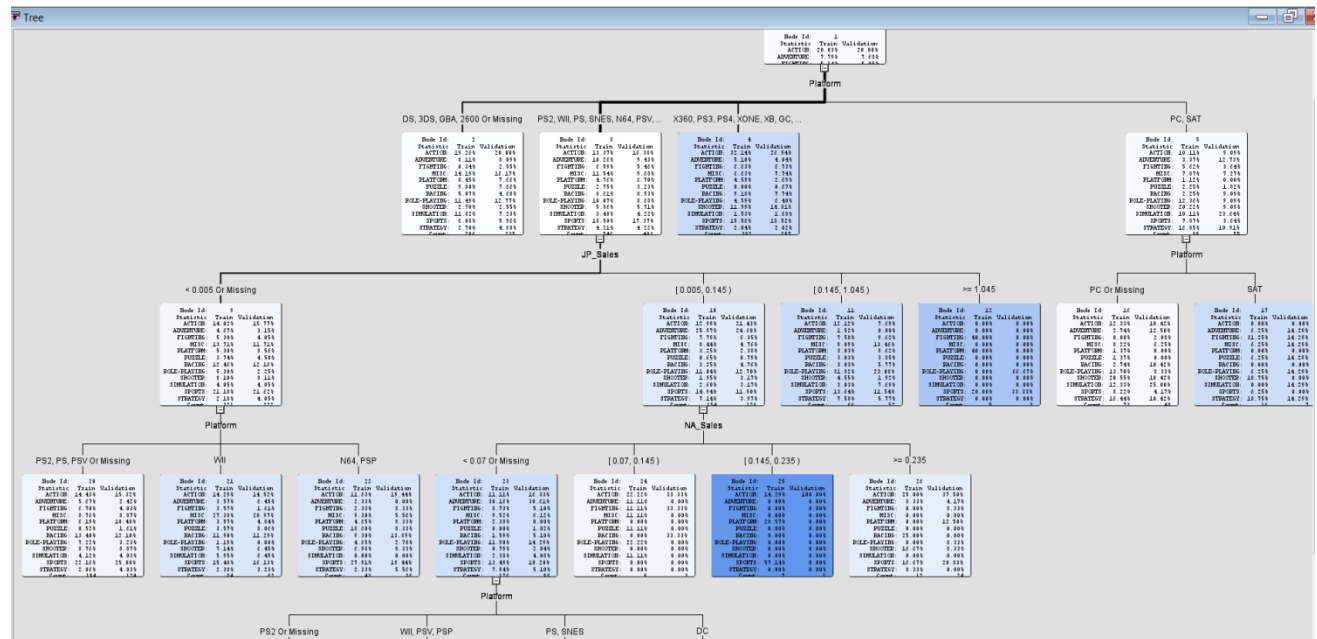
Fit Statistics

Target	Target Label	Fit Statistics	Statistics Label	Train	Validation	Test
Genre		NOBS	Sum of Freq...	1323	990	1006
Genre		MISC	Misclassifica...	0.73167	0.761616	0.776342
Genre		MAX	Maximum Ab...	0.996622	1	1
Genre		SSE	Sum of Squa...	1122.579	867.1573	894.7488
Genre		ASE	Average Squ...	0.070709	0.072993	0.074118
Genre		RASE	Root Averag...	0.265912	0.270172	0.272246
Genre		DIV	Divisor for A...	15876	11880	12072
Genre		DFT	Total Degree...	14553		

Output

102	2	1	296	0.03	235	0.05
103	20	3	194	0.02	124	0.04
104	21	3	84	0.02	62	0.03
105	16	2	73	0.16	48	0.10
106	28	4	71	0.08	53	0.04
107	11	2	66	0.08	52	0.06
108	22	3	43	0.02	36	0.06
109	27	4	31	0.10	30	0.07
110	29	4	17	0.06	14	0.07
111	17	2	16	0.19	7	0.14
112	26	2	12	0.08	24	0.08

Analyzing more on the accuracy of Decision tree, B4D4, we did not consider baseline misclassification taught in class as our data set deals with multiple classes rather than just two classes. However, we do look at the misclassification rate between train, validate, and test, and the difference between them does not exceed 5% so it is still good.



This is the tree node from the results window. We can see that the genre of games are divided first based on the platform or console. Some consoles support certain type of genre more than others (such as PC with simulations as game controllers are harder to play simulation games). In the second level of the tree is where it gets more interesting. The genres are further divided into sales after JP region, and since further division of NA region sales occurs in the lower sales of JP region, it tells us some genres favored in JP region are different for those that are favored in NA region. For all consoles, one exception is for pc. It makes sense that many genres are available on pc as it is compatible to play all types of genres, and games available on PC are not affected much by region. As stated, before in tableau analysis, JP region may prefer handheld devices due to smaller spaces in living, while NA favors larger and more powerful consoles. This exception of console applies to PCs as every household would usually own one PC. So, the genre of games available on PCs are not affected either by the limitation of console due to region, nor the limitation of gameplay due to the nature of console. This is one reason why many games that are produced in different platforms are almost always available on PCs.

For exploratory we have done stability testing of clusters by checking whether the clusters before and after sampling are within the ± 2 range. Based on the 90% sample, we obtained 22 clusters for the 90% ward clustering, 10 clusters for the 90% centroid clustering, and 5 clusters for the 90% average clustering. From this information, we can conclude that the number of clusters before and after sampling remains within a range of ± 2 clusters for all the clustering algorithms, including ward, centroid, and average clustering. We have done the same for user

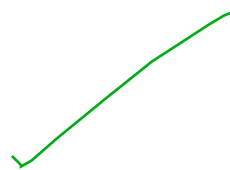
specified and found the clusters to be stable.

Additionally, discussing the information presented by stable clusters will add further rigor in identifying

Scoring the model:

We will now compare our model with Score data to see the accuracy of the model. In the excel sheet generated from scoring the data, the genre is predicted against EM_Classification, and our results shows out of 3319 total observations, our model predicted 816 times correctly. The accuracy of our model is approximately at 24.6%. ("Datacadamia", 2020)

Total	3319
Wrong	2503
Right	816
% Accuracy	24.6%



Is the model good or bad? 24.6% is less than a toss of a coin for a hit or miss, but to predict whether a game falling under a certain genre will perform well in sales in a given region, is not as simple as heads or tails. There are more than two genre (in fact more than 6). The limitation of the model, and the dataset also applies, in that some games can have more than one categories, but is not accounted for in the data set.

Conclusion and practical (actionable) recommendations based:

As we have discussed in the report, the data has wider range implications than just pertaining to video game industry. The application of the dataset can be extended to psychology and sociology. The type of games a society plays also indicates what the society is going through. But the main recommendation, In the analysis, we observe that games are categorized based on the platform or console, where certain genres are more prevalent on specific platforms. For example, simulations are more common on PC due to the availability of game controllers that are well-suited for such games. Moving further down the tree, the genres are divided based on sales in regions such as JP (Japan) and NA (North America). The subdivision of NA region sales occurs primarily in lower sales categories of JP region, indicating that genre preferences differ between JP and NA regions. However, there is an exception for PCs. PCs offer a wide range of genres as they are compatible with various game types, and their availability is not significantly influenced by region. As mentioned earlier in the tableau analysis, JP region tends to prefer handheld devices due to limited living spaces, while NA region (With typically bigger houses) favors larger and more powerful consoles. Nevertheless, this exception does not apply to PCs



since they are commonly owned by households. Consequently, the genre of games available on PCs is not constrained by regional limitations or gameplay restrictions associated with specific consoles. This flexibility is a key factor in why games produced for different platforms are often available on PCs.

Summary & Phase of Team Contributions:

Milestone	Activities
Phase 1: Proposal Completed: 4/29/2023	Identify the problem that needs to be solved from our selected dataset. Once the Research Questions are done, we drafted methodologies to tackle the issue, develop steps and milestones, and submit this proposal.
Phase 2: Data Preparation Completed: 5/2/2023 Data Set Selection: Tanmai Reddy Data Preparation: Nyan, Tanmai, and Rahul	In order to begin our analysis, we will begin cleaning our data, and making sure it is ready to be processed by the team. This is set to be done the following week, with 5/2/2023 as the latest.
Phase 3: Model Development, Evaluation & Deployment Completed: 5/10/2023 Tableau Visualization & Analysis: Nyan and Tanmai Literature Review: Tanmai, Rahul, and Dinesh Exploratory Modeling & Analysis: Nyan and Tanmai Predictive Modeling & Analysis: Nyan and Tanmai Abstract & Report Methodology: Rahul, Dinesh, and Nyan Scoring: Tanmai Recommendations: Nyan	We will be using exploratory analysis with different classification methods to generate a model from our game sales dataset (both supervised and unsupervised maybe used). The best model will then be selected through sampling into 3 data sets (train, validate, test), and with score data set. We will then deploy our model to answer our research questions using supervised model based on the model generated from the exploratory analysis. 5/5/2023 will be the targeted completion date

Phase 4: Report Compilation Complete by: 5/14/2023	We will interpret the results and explain the analysis in detail compiling it in a team report.
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References

Bradsher, K. (2003). *High and mighty: The dangerous rise of the SUV*. Public Affairs.

K. P. Sinaga and M. -S. Yang, "*Unsupervised K-Means Clustering Algorithm*," in IEEE Access, vol. 8, pp. 80716-80727, 2020, doi: 10.1109/ACCESS.2020.2988796.

Song, Y.-Y., & Lu, Y. (2015, April 25). *Decision tree methods: Applications for classification and prediction*. Shanghai archives of psychiatry.
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4466856/>

Data Mining - scoring (applying). Datacadamia. (2020, March 23).
https://datacadamia.com/data_mining/scoring

(PDF) the impact of platform on global video game sales - researchgate. (n.d.). Retrieved April 30, 2023, from
https://www.researchgate.net/publication/297754899_The_Impact_Of_Platform_On_Global_Video_Game_Sales